NEET UG 2027: Chemistry Formula Compendium

Module 01: Mole Concept, Atomic Structure, Thermodynamics & Equilibrium

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Strict NCERT & NMC Alignment | Complete Mathematical Variations, Use-Cases & Traps

1 Some Basic Concepts in Chemistry: Mole Concept (Unit 1)

1.1 Stoichiometric Mole Formulations

Fundamental Mole Formulations & Gas Volumes

$$n = \frac{w \text{ [mass in g]}}{M \text{ [molar mass in g mol}^{-1}\text{]}} = \frac{N \text{ [number of particles]}}{N_A} = \frac{V_{\text{STP}} \text{ [Volume of ideal gas at}}{22.4\,\text{L}} \tag{1}$$

$$N_A = 6.022 \times 10^{23}\,\text{mol}^{-1}, \qquad 1\,\text{amu} = 1\,\text{u} = \frac{1}{N_A}\,\text{g} \approx 1.66 \times 10^{-24}\,\text{g} = 1.66 \times 10^{-27}\,\text{kg} \tag{2}$$

$$\text{Vapour Density (VD)} = \frac{\text{Density of gas at } T, P}{\text{Density of } H_2 \text{ gas at same } T, P} \implies \text{Molar Mass} = 2 \times \text{VD} \tag{3}$$

$$\text{Mass \% of element } X = \frac{(\text{Number of atoms of } X) \times (\text{Atomic mass of } X)}{\text{Molar mass of compound}} \times 100 \tag{4}$$

Limiting Reagent Condition for Reaction $aA + bB \rightarrow cC + dD$**:**

$$\text{Compare ratio } \frac{n_A}{a} \text{ and } \frac{n_B}{b} \implies \text{If } \frac{n_A}{a} < \frac{n_B}{b}, \quad A \text{ is the limiting reagent (completely consumed).} \tag{5}$$

1.2 Solution Concentrations & Dilution Formulations

Molarity, Molality, Normality & Interconversions

$$M = \frac{n_{\text{solute}}}{V_{\text{solution}}\ (\text{in L})} = \frac{w_B \times 1000}{M_B \times V_{\text{sol}}\ (\text{in mL})} \quad \text{[Molarity: Temperature Dependent]} \tag{6}$$

$$m = \frac{n_{\text{solute}}}{W_{\text{solvent}}\ (\text{in kg})} = \frac{w_B \times 1000}{M_B \times W_A\ (\text{in g})} \quad \text{[Molality: Temperature Independent]} \tag{7}$$

$$N = \frac{\text{Gram equivalents of solute}}{V_{\text{solution}}\ (\text{in L})} = M \times n\text{-factor} \quad \text{[Normality: } n\text{-factor is valence/redox exch} \tag{8}$$

$$x_B = \frac{n_B}{n_A + n_B}, \qquad x_A + x_B = 1 \quad \text{[Mole Fraction: Temperature Independent} \tag{9}$$

$$\text{ppm} = \frac{\text{Mass of component}}{\text{Total mass of solution}} \times 10^6, \qquad \%(\text{w/w}) = \frac{w_{\text{solute}}}{w_{\text{solution}}} \times 100 \tag{10}$$

High-Yield Interconversion Formulas (d = solution density in g mL^{-1}):

$$M = \frac{10 \times \%(w/w) \times d}{M_B}, \qquad m = \frac{1000M}{1000d - MM_B}, \qquad m = \frac{x_B \times 1000}{(1 - x_B)M_A} \tag{11}$$

Dilution and Mixing of Solutions:

$$M_1V_1 = M_2V_2, \qquad M_{\text{mix}} = \frac{M_1V_1 + M_2V_2 + \ldots}{V_1 + V_2 + \ldots}, \qquad N_1V_1 = N_2V_2 \quad \text{[at equivalence point]} \tag{12}$$

Mole Traps: STP Standards & Temperature Dependence

- **STP Standard Volume:** In IUPAC revised standard conditions ($T = 273.15\,\text{K}, P = 1\,\text{bar} = 10^5\,\text{Pa}$), molar volume is 22.7 L. Under traditional NCERT standard pressure ($1\,\text{atm} = 1.01325 \times 10^5\,\text{Pa}$), molar volume is 22.4 L. Unless specified as 1 bar, calculate using 22.4 L.
- **Molarity Changes with Temperature:** Because volume expands on heating ($V \propto T$), molarity M and normality N **decrease with increasing temperature**. Molality m, mole fraction x, and $\%(w/w)$ involve mass only and remain **strictly constant**.

2 Structure of Atom (Unit 2)

2.1 Quantum Mechanics, Wave-Particle Duality & Uncertainty

de Broglie, Heisenberg Uncertainty & Orbital Angular Momentum

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \qquad [h = 6.626 \times 10^{-34}\,\text{J s}] \tag{13}$$

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \implies \Delta x \cdot (m\Delta v) \geq \frac{h}{4\pi} \implies \Delta x \cdot \Delta v \geq \frac{h}{4\pi m} \qquad \text{[Heisenberg Uncertainty Principle]} \tag{14}$$

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi} \qquad \text{[Energy-Time Uncertainty Formulation]} \tag{15}$$

$$L_{\text{orbital}} = \sqrt{l(l+1)}\hbar = \sqrt{l(l+1)}\frac{h}{2\pi} \qquad \text{[Orbital Angular Momentum; } l = 0 \text{ for } s\text{-orbital!]} \tag{16}$$

$$\mu_{\text{spin}} = \sqrt{n(n+2)}\,\text{BM} \quad [\text{BM} = \text{Bohr Magneton}] \qquad [n = \text{number of unpaired electrons}] \tag{17}$$

2.2 Quantum Numbers, Nodes & Electron Filling Principles

Quantum Number	Symbol	Allowed Values	Physical Significance / Property
Principal	n	$1, 2, 3, \ldots, \infty$	Main energy level, shell size, distance from nucleus
Azimuthal / Orbital Angular	l	$0, 1, 2, \ldots, (n-1)$	Subshell shape ($l = 0 : s, 1 : p, 2 : d, 3 : f$)
Magnetic	m_l	$-l, \ldots, 0, \ldots, +l$	Spatial orientation of orbital in magnetic field
Spin	m_s	$+\frac{1}{2}, -\frac{1}{2}$	Intrinsic spin angular momentum orientation

Table 1: The Four Quantum Numbers and Boundary Constraints.

Nodal Formulations for Atomic Orbitals

$$\text{Radial Nodes (Spherical Nodes)} = n - l - 1 \tag{18}$$

$$\text{Angular Nodes (Nodal Planes)} = l \tag{19}$$

$$\text{Total Nodes} = (n - l - 1) + l = n - 1 \tag{20}$$

$$\text{Maximum Electrons in Shell } n = 2n^2, \qquad \text{Maximum Orbitals in Shell } n = n^2 \tag{21}$$

$$\text{Maximum Electrons in Subshell } l = 2(2l + 1) = 4l + 2 \tag{22}$$

Orbital Filling Rules & Exchange Energy Stability

- **Aufbau Principle ($(n + l)$ Rule):** Electrons enter lowest energy subshell first. Lower $(n + l)$ has lower energy. If $(n + l)$ identical, lower n has lower energy (e.g. $4s$ has $4 + 0 = 4$, $3d$ has $3 + 2 = 5 \implies 4s$ fills before $3d$).
- **Pauli Exclusion Principle:** No two electrons in an atom can have the identical set of all four

quantum numbers. An orbital holds maximum 2 electrons with opposite spins.

- **Hund's Rule of Maximum Multiplicity:** Pairing does not occur in degenerate orbitals until each holds one electron with parallel spin.
- **Exchange Energy of Half/Fully Filled Subshells:** Number of possible electron exchanges among electrons of parallel spin:

$$K = \frac{n_p(n_p - 1)}{2} \implies \text{For } d^5 : K = \frac{5 \times 4}{2} = 10 \text{ exchanges (extraordinarily stable configuration!)}$$

3 Chemical Thermodynamics & Thermochemistry (Unit 4)

3.1 First Law, Work, Enthalpy & Heat Capacities

First Law, PV-Work & Enthalpy-Internal Energy Coupling

$$\Delta U = q + w \quad \text{[IUPAC Sign: Work done ON system is positive; BY system is negative]} \tag{23}$$

$$w_{\text{irr}} = -P_{\text{ext}}(V_2 - V_1) = -P_{\text{ext}}\Delta V \quad \text{[Irreversible} \tag{24}$$

$$w_{\text{rev, isothermal}} = -\int_{V_1}^{V_2} P\,dV = -nRT\ln\left(\frac{V_2}{V_1}\right) = -2.303nRT\log_{10}\left(\frac{P_1}{P_2}\right) \tag{25}$$

$$w_{\text{free expansion}} = 0 \quad (P_{\text{ext}} = 0 \text{ into vacuum}) \implies \Delta U = 0, \quad \Delta T = 0 \text{ for ideal gas} \tag{26}$$

$$H = U + PV \implies \Delta H = \Delta U + \Delta(PV) = \Delta U + \Delta n_g RT \quad [\Delta n_g = \sum n \tag{27}$$

$$C_p - C_v = R \implies \Delta H = q_p = nC_p\Delta T, \qquad \Delta U = q_v = nC_v\Delta T \tag{28}$$

3.2 Thermochemistry & Hess's Law

Hess's Law, Reaction Enthalpy & Bond Dissociation Enthalpies

$$\Delta_r H^\circ = \sum \Delta_f H^\circ(\text{Products}) - \sum \Delta_f H^\circ(\text{Reactants}) \quad [\Delta_f H^\circ \text{ of pure elements in standard state is 0}] \tag{29}$$

$$\Delta_r H^\circ = \sum BE(\text{Reactants}) - \sum BE(\text{Products}) \quad \text{[All species MUST be in GASEOUS state!]} \tag{30}$$

$$\Delta_r H = \sum \Delta_{\text{comb}} H^\circ(\text{Reactants}) - \sum \Delta_{\text{comb}} H^\circ(\text{Products}) \quad \text{[From Enthalpies of Combustion]} \tag{31}$$

Kirchhoff's Law of Temperature Dependence of Enthalpy:

$$\frac{\Delta H_{T_2} - \Delta H_{T_1}}{T_2 - T_1} = \Delta C_p, \qquad \frac{\Delta U_{T_2} - \Delta U_{T_1}}{T_2 - T_1} = \Delta C_v \quad \left[\Delta C_p = \sum C_{p,\text{prod}} - \sum C_{p,\text{react}}\right] \tag{32}$$

3.3 Entropy, Second Law & Gibbs Free Energy Spontaneity

Entropy, Third Law & Spontaneity Criteria

$$\Delta S = \frac{q_{\text{rev}}}{T}, \qquad \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0 \quad \text{[Second Law]} \tag{33}$$

$$\Delta S_{\text{system}} = nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right) = nC_p \ln\left(\frac{T_2}{T_1}\right) - nR \ln\left(\frac{P_2}{P_1}\right) \tag{34}$$

$$\Delta S_{\text{isothermal}} = nR \ln\left(\frac{V_2}{V_1}\right) = 2.303nR \log_{10}\left(\frac{P_1}{P_2}\right), \qquad \Delta S_{\text{isochoric}} = nC_v \ln\left(\frac{T_2}{T_1}\right) \tag{35}$$

$$G = H - TS \implies \Delta G = \Delta H - T\Delta S \quad \text{[Gibbs-Helmholtz Equation at constant } T, P\text{]} \tag{36}$$

$$\Delta G^\circ = -RT \ln K = -2.303RT \log_{10} K \iff K = e^{-\Delta G^\circ / RT} \tag{37}$$

ΔH	ΔS	$\Delta G = \Delta H - T\Delta S$	**Spontaneity Criterion**
$-$	$+$	Always Negative $(-)$	Spontaneous at ALL temperatures
$+$	$-$	Always Positive $(+)$	Non-spontaneous at ALL temperatures (reverse is spontaneous)
$-$	$-$	Negative at low T $(T < \frac{\Delta H}{\Delta S})$	Spontaneous at LOW temperature; non-spontaneous at high T
$+$	$+$	Negative at high T $(T > \frac{\Delta H}{\Delta S})$	Spontaneous at HIGH temperature; non-spontaneous at low T

Table 2: Spontaneity Matrix Governed by Enthalpy and Entropy Interplay.

Thermodynamics Traps: Standard States & Bond Enthalpy Reversal

- **Bond Enthalpy Subtraction Order:** Enthalpy of reaction using formation enthalpies is [Products − Reactants], but using bond enthalpies it is [**Reactants − Products**]!
- **Reference Standard State Formation Enthalpy is Zero:** Only the most stable allotropic form at 298.15 K and 1 bar has $\Delta_f H^\circ = 0$: Carbon is **Graphite** (Diamond has $\Delta_f H^\circ = 1.9\,\text{kJ mol}^{-1}$), Sulfur is **Rhombic**, Phosphorus is **White**.

4 Equilibrium (Unit 6)

4.1 Chemical Equilibrium, $K_p - K_c$ Coupling & Le Chatelier

Equilibrium Constants, Reaction Quotient & Degree of Dissociation

$$K_p = K_c(RT)^{\Delta n_g} \quad \left[R = 0.0821\,\text{L atm mol}^{-1}\,\text{K}^{-1} \text{ or } 0.0831\,\text{L bar mol}^{-1}\,\text{K}^{-1}\right] \tag{38}$$

$$Q < K \implies \text{Forward direction}; \quad Q = K \implies \text{Equilibrium}; \quad Q > K \implies \text{Backward direction} \tag{39}$$

$$\alpha = \frac{D-d}{(n-1)d} = \frac{M_{\text{theoretical}} - M_{\text{observed}}}{(n-1)M_{\text{observed}}} \quad [n = \text{number of moles produced per mole dissociated}] \tag{40}$$

van 't Hoff Equation (Temperature Dependence of Equilibrium Constant):

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \implies \log_{10}\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] \tag{41}$$

4.2 Ionic Equilibrium: Ostwald Dilution, Water Autoionization & pH

Ostwald's Dilution Law, Water Ion Product & Acid-Base pH

$$K_a = \frac{C\alpha^2}{1-\alpha} \approx C\alpha^2 \implies \alpha = \sqrt{\frac{K_a}{C}}, \quad [\text{H}^+] = C\alpha = \sqrt{K_a C} \quad (\text{for } \alpha \le 0.05) \tag{42}$$

$$K_w = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14} \text{ at } 25^\circ\text{C} \implies \text{pH} + \text{pOH} = 14 = \text{p}K_w \tag{43}$$

$$\text{pH} = -\log_{10}[\text{H}^+], \qquad [\text{H}^+] = 10^{-\text{pH}} \tag{44}$$

4.3 Salt Hydrolysis Formulations

Salt Type (Example)	Hydrolysis Const. (K_h)	Degree of Hydrolysis (h)	pH Formula at 25°C
Strong Acid + Strong Base (NaCl)	No Hydrolysis	$h = 0$	pH = 7.00
Weak Acid + Strong Base (CH_3COONa)	$K_h = \frac{K_w}{K_a}$	$h = \sqrt{\frac{K_w}{CK_a}}$	pH = $7 + \frac{1}{2}\text{p}K_a + \frac{1}{2}\log C$
Strong Acid + Weak Base (NH_4Cl)	$K_h = \frac{K_w}{K_b}$	$h = \sqrt{\frac{K_w}{CK_b}}$	pH = $7 - \frac{1}{2}\text{p}K_b - \frac{1}{2}\log C$
Weak Acid + Weak Base (CH_3COONH_4)	$K_h = \frac{K_w}{K_a K_b}$	$h = \sqrt{\frac{K_w}{K_a K_b}}$	pH = $7 + \frac{1}{2}\text{p}K_a - \frac{1}{2}\text{p}K_b$

Table 3: Exact Hydrolysis Parameters and pH Formulations for Canonical Salts.

4.4 Buffer Solutions & Henderson-Hasselbalch Equations

Acidic and Basic Buffer Governing Equations

$$\text{pH} = \text{p}K_a + \log_{10}\left(\frac{[\text{Salt}]}{[\text{Weak Acid}]}\right) = \text{p}K_a + \log_{10}\left(\frac{[\text{Conjugate Base}]}{[\text{Acid}]}\right) \quad \text{[Acidic Buffer]} \tag{45}$$

$$\text{pOH} = \text{p}K_b + \log_{10}\left(\frac{[\text{Salt}]}{[\text{Weak Base}]}\right) \implies \text{pH} = 14 - \text{pOH} \quad \text{[Basic Buffer]} \tag{46}$$

$$\text{Buffer Range} = \text{p}K_a \pm 1, \qquad \beta = \frac{dB}{d(\text{pH})} \quad \text{[Buffer Capacity } \beta \text{ is maximu} \tag{47}$$

4.5 Solubility Product (K_{sp}) & Precipitation Criteria

Solubility Product Equations & Precipitation

$$A_xB_y(s) \rightleftharpoons xA^{y+}(aq) + yB^{x-}(aq) \implies K_{sp} = [A^{y+}]^x[B^{x-}]^y = (xS)^x(yS)^y = x^x y^y S^{x+y} \tag{48}$$

Canonical Stoichiometric Solubility Forms:

- $1:1$ Salt ($AgCl$): $K_{sp} = S^2 \implies S = \sqrt{K_{sp}}$
- $1:2$ or $2:1$ Salt ($PbCl_2, CaF_2, Ag_2CrO_4$): $K_{sp} = 1^1 \times 2^2 \times S^{1+2} = 4S^3 \implies S = \sqrt[3]{\frac{K_{sp}}{4}}$
- $1:3$ or $3:1$ Salt ($Al(OH)_3, Fe(OH)_3$): $K_{sp} = 1^1 \times 3^3 \times S^{1+3} = 27S^4 \implies S = \sqrt[4]{\frac{K_{sp}}{27}}$
- $2:3$ or $3:2$ Salt ($Ca_3(PO_4)_2, As_2S_3$): $K_{sp} = 2^2 \times 3^3 \times S^{2+3} = 108S^5 \implies S = \sqrt[5]{\frac{K_{sp}}{108}}$

Precipitation Criterion:

$$Q_{sp} < K_{sp} \implies \text{Unsaturated (No precipitate)}; \quad Q_{sp} = K_{sp} \implies \text{Saturated}; \quad Q_{sp} > K_{sp} \implies \text{Precipitation Oc} \tag{49}$$

Ionic Equilibrium Traps: Autoionization of Water & pH Scaling

- **Low Concentration Acid pH (10^{-8} M HCl):** Never write pH = 8 for an acid! In extremely dilute solutions ($< 10^{-6}$ M), autoionization of water ($[H^+]_{water} = 10^{-7}$ M) cannot be neglected:

$$[H^+]_{total} = 10^{-8} + 10^{-7} = 1.1 \times 10^{-7}\,\text{M} \implies \text{pH} \approx 6.96 \quad (\text{strictly } < 7)$$

- **Temperature Dependence of K_w:** The autoionization of water ($H_2O \rightleftharpoons H^+ + OH^-$) is endothermic ($\Delta H > 0$). Heating water to 60°C increases $K_w \approx 10^{-13}$. Neutral water at 60°C has pH $= \frac{1}{2}\text{p}K_w = 6.5$, but remains strictly **neutral** because $[H^+] = [OH^-]$.

End of Module 01 Physical Chemistry Formula Handbook

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